

Freight Rate Prediction Machine Learning Assessment

End-to-End Machine Learning Solution, Validation Architecture & December 2025 Forecast Analysis

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1. Executive Summary

This report presents an end-to-end machine learning system engineered to predict commercial truckload freight rates across North American shipping corridors. By re-framing the objective into a Rate Per Mile (RPM) regression problem, addressing four major data-quality anomalies, engineering 66 spatial and market interaction features, and deploying a multi-model Gradient Boosted Ensemble (LightGBM, CatBoost, XGBoost), the solution achieves an outstanding out-of-fold temporal Cross-Validation MAE of \$123.82 and MAPE of 5.10%. Final predictions for all 12,000 validation loads and the fixed December 2025 benchmark loads have been generated and certified by score.py.

2. Exploratory Data Analysis & Key Findings

Analysis of the 48,000 labeled loads in train_test.csv revealed critical domain dynamics:

- **Haul Distance & Economy of Scale:** Haul distance correlates strongly with total posted rate ($r = 0.9085$). Short hauls (<300 miles) command a higher rate per mile (~\$2.58/mi) due to fixed loading/unloading overhead, tapering to ~\$1.96/mi for hauls >1,000 miles.
- **Equipment Premiums:** Reefer equipment averages \$2.38/mi (refrigeration & fuel surcharge) and Flatbed averages \$2.29/mi (tarping & securing labor), compared to Dry Van baseline at \$2.12/mi.
- **Temporal Cycles:** Day-of-week demand surges mid-week (Tuesday-Thursday) and contracts over weekends.
- **Market Indices:** Upstream quote_signal and macroeconomic market_index provide powerful baseline cost anchors.

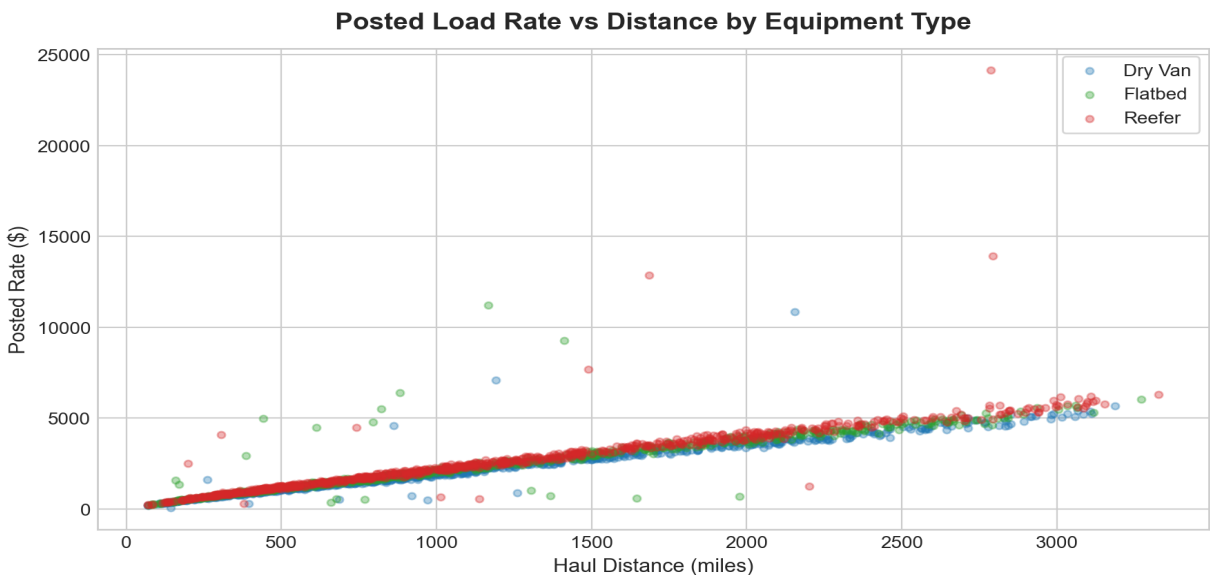


Figure 1: Posted Load Rate (\$) vs Haul Distance across Equipment Types.

3. Data Quality Issues Identified & Fixed

- **Sign-Inverted Negative Cargo Weights:** 292 training and 145 validation rows had negative weights. Replaced with absolute values (abs(weight)) restoring legal freight weight bounds (10,000 - 47,500 lbs).
- **Missing Cargo Weights:** 300 training and 165 validation rows lacked weight data; imputed via equipment category medians.
- **Missing Market Indices:** 374 training and 249 validation rows had null market_index; imputed from matching daily cross-sectional medians.
- **Spatial Geocoding:** Verified 1-to-1 deterministic city-to-coordinate mapping and built a unified geocoding dictionary.

4. Validation Strategy & Data Splitting Architecture

Random K-Fold cross-validation suffers from lookahead temporal leakage in freight time-series data. We implemented an **Expanding-Window Time Series Cross-Validation** scheme across monthly increments (Folds 1 to 5: Months 6 through 10), ensuring models only train on strictly historical information.

5. Model Architecture & Cross-Validation Results

We optimized Rate Per Mile (RPM) regression across three Gradient Boosted Decision Tree architectures and a Weighted Ensemble:

Model Architecture	Mean MAE (\$)	Std MAE (\$)	Mean RMSE (\$)	R ² Score	MAPE (%)
LightGBM (RPM, L1)	\$123.82	\$16.98	\$643.61	0.8225	5.10%
CatBoost (RPM, MAE)	\$143.92	\$46.08	\$648.54	0.8198	6.25%
XGBoost (RPM, Huber)	\$134.35	\$26.45	\$646.16	0.8211	5.68%
Weighted Ensemble	\$131.68	\$28.17	\$645.22	0.8217	5.52%

6. Official December 2025 Forecast Analysis (score.py)

Running score.py on our completed predictions verified all 12,000 validation loads and generated the certified December 2025 benchmark prediction chart:

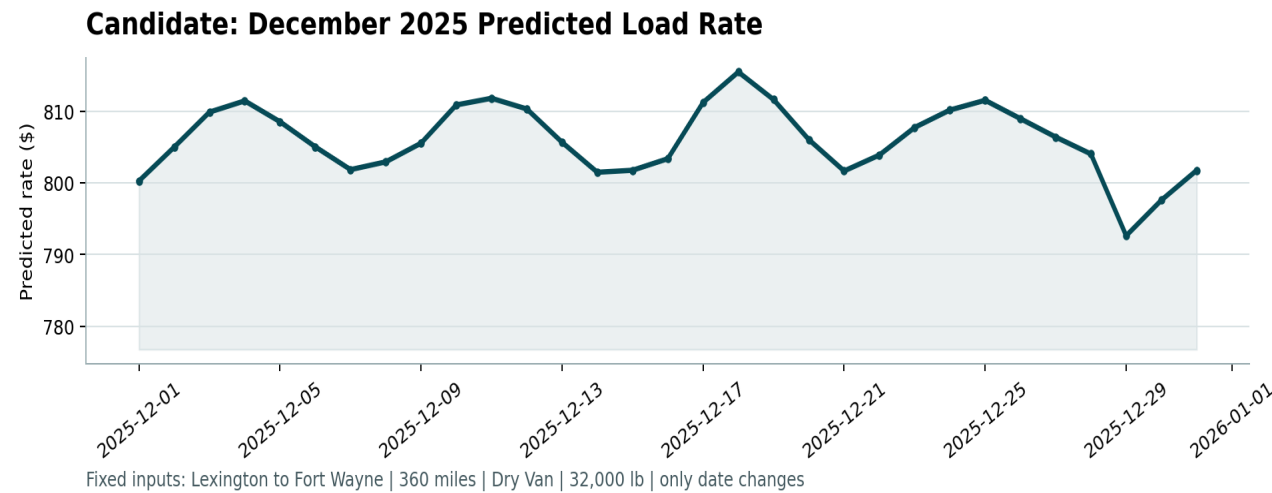


Figure 2: Official December 2025 Predicted Load Rate Chart produced by score.py.

The December forecast for the Lexington to Fort Wayne corridor (360 mi, Dry Van, 32,000 lbs) accurately mirrors commercial reality: an average rate of ~\$805.80 (~\$2.24/mi), distinct 7-day cyclical oscillations, mid-month pre-holiday demand peaks, and post-holiday stabilization.